

DATABASE SYSTEMS

Chapter 1 — Explained in Simple Words

Based on: Coronel & Morris, Database Systems: Design, Implementation & Management, 13th Ed.

Data vs. Information

What's the difference?

D DATA

Plain, raw facts — like a number or a name

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- It has no meaning yet, until it's organized
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- It is the raw material used to build information
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- Data management means collecting, storing, and finding data

I INFORMATION

Made by processing (organizing) the data

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- It shows us the real meaning behind the data
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- It helps us build new knowledge
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- Good information must be correct, useful, and on time

What is a Database?

And what is a DBMS?

DATABASE

A shared place that stores related data together

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- It keeps end-user data — the real facts people need
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- It also keeps metadata — data that describes and connects the other data

DBMS

(Database Management System)

A group of programs that manages the database

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- It controls the structure of the database
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- It controls who can access the data and how

What Does a DBMS Do?

It sits between the user and the database

- 1 It works as a middleman between the user and the database
- 2 It lets many people share the same data
- 3 It gives users one simple, combined view of the data
- 4 It takes app requests and turns them into database actions
- 5 It hides the complicated inner workings from the user

Advantages of Using a DBMS

Why is it better than storing data in plain files?

1 Better data sharing

Users get easy, well-managed access to the data they need

2 Better data security

The DBMS helps protect data through proper access rules

3 Better data integration

All the data is linked, giving a clear overall picture

4 Less inconsistency

Same data isn't stored many different ways in many places

5 Faster answers

Users can quickly ask questions and get results (queries)

6 Higher productivity

End users can do more work in less time

Types of Databases (1)

Grouped by number of users and location

Single-user

Used by one person at a time (like a desktop database)

Multouser

Used by many people at the same time

Workgroup

Used by a small team or one department

Enterprise

Used by many departments across the whole company

Centralized

All the data is kept in one single location

Distributed

Data is spread across several different locations

Cloud

Built and run using online cloud services

Types of Databases (2)

Grouped by purpose and by data structure

BY PURPOSE

General-purpose: holds many kinds of data for many uses

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- Discipline-specific: focused on one subject area
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- Operational: runs the company's daily work
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- Analytical: stores history and numbers for making decisions
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- Data warehouse: organizes data to support decisions

BY DATA STRUCTURE

OLAP: lets users explore data from the warehouse

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- Business intelligence: turns business data into useful insight
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- Unstructured data: kept in its original, unorganized form
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- Structured data: has been arranged into a clear format
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- Semi-structured data: partly organized (example: XML)

Database Design

Getting the structure right matters a lot

Database design means planning the structure that will store and manage the data

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- A well-designed database makes data easy to manage and gives accurate, useful information
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- A poorly-designed database causes errors that are hard to find and fix

HOW FILE SYSTEMS EVOLVED OVER TIME



Basic File Terms

The building blocks of a simple file system

Term	Simple Meaning
Data	Raw facts, such as a phone number or a name. Alone, they don't mean much.
Field	A single piece of data, like just the 'name' or just the 'phone number'
Record	A group of related fields — for example, one customer's full details
File	A collection of related records — for example, all customers together

Problems With Old File Systems

Why companies moved away from plain file storage

- 1 Took a long time to build and set up
- 2 Hard to get quick answers to new questions
- 3 Complicated to manage and maintain
- 4 Weak security and hard to share data safely
- 5 Needed a lot of custom programming

Dependence & Data Redundancy

Two more problems old file systems had

STRUCTURAL / DATA DEPENDENCE

If the file's structure changes, every program using it must also change

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- This is called structural dependence
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- Data dependence: the way we access data changes when its storage details change
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- A good system keeps the program working even if storage details change (data independence)

DATA REDUNDANCY

This means storing the same data in more than one place without needing to

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- It creates scattered 'islands' of data
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- It can lead to a data anomaly — when a change is made in one place but not everywhere it should be
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- Three kinds: update, insertion, and deletion anomalies

The 5 Parts of a Database System

Hardware, software, people, procedures, and data

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Hardware

All the physical devices — computers, servers, storage, and network equipment

2

Software

The operating system, the DBMS itself, and the apps that use the data

3

People

System admins, database admins (DBAs), designers, programmers, and end users

4

Procedures

The rules and instructions for how the system should be designed and used

5

Data

The actual facts stored in the database — the raw material for information

What Jobs Does a DBMS Do?

Its main functions, explained simply

Data dictionary

Keeps a record of what each piece of data means

Storage management

Keeps the database running fast and efficiently

Data transformation

Changes entered data into the required format

Security

Controls who can see and use which data

Multiuser access

Lets many people use the data at once, safely

Backup & recovery

Helps restore the database if something goes wrong

Data integrity

Keeps data accurate and avoids repeated copies

Query language (SQL)

Lets users ask for data without saying exactly how to get it

Database Career Opportunities

Jobs you can do with database skills

Job Title	What They Do
Database Developer	Builds and maintains apps that use databases
Database Designer	Designs and maintains the database structure
Database Administrator	Manages and looks after the DBMS and database
Database Analyst	Builds databases for reports and decision-making
Database Architect	Plans the full database setup, start to finish
Database Consultant	Helps companies use databases to work better
Database Security Officer	Sets up and manages data security rules
Cloud Data Architect	Builds databases that run in the cloud

KEY TAKEAWAY

A database, managed well by a DBMS, turns scattered raw data into shared, accurate, and easy-to-use information.